

Local Mertens Energy: Brownian-Type Kernel Spectral Decomposition and a Conditional Limiting Distribution

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Abstract

We study the local quadratic energy

$$\mathcal{M}_L(X) = \int_{e^{-L}X}^X \frac{|M(y) - M(e^{-L}X)|^2}{y^2} dy, \quad M(x) = \sum_{n \leq x} \mu(n).$$

The first part gives an exact spectral decomposition of this energy through the Brownian-type kernel

$$K_L(s, t) = \frac{1}{\max(s, t)} - 1$$

on $[e^{-L}, 1]$, a time-changed Brownian covariance. The second part is conditional: assuming the Riemann Hypothesis and the Gonek–Hejhal type bound $J_{-1}(T) = \sum_{0 < \gamma \leq T} |\zeta'(\rho)|^{-2} \ll T$, we prove that $Y \mapsto \mathcal{M}_L(e^Y)$ possesses a limiting distribution and identify its first moment. Under the additional hypothesis of linear independence of the zero ordinates, the limit law is that of an explicit random Hilbert-space series over the nontrivial zeros of ζ . We also record a quantitative lower bound on the oscillation of $\mathcal{M}_L(e^Y)$ under the weaker assumption that γ_1 is rationally independent of the remaining ordinates. This extends the limiting-distribution results of Ng for the linear functional $e^{-y/2}M(e^y)$ to a quadratic functional of a fixed-ratio window, and is a Möbius analogue of the fixed-ratio mean-square problem for the prime number theorem studied by Brent, Platt and Trudgian. Numerical computations up to $X = 10^8$ with the first 1000 zeros are in agreement with the explicit-formula description.

1 Introduction

We study the local quadratic energy

$$\mathcal{M}_L(X) = \int_{e^{-L}X}^X \frac{|M(y) - M(e^{-L}X)|^2}{y^2} dy, \quad M(x) = \sum_{n \leq x} \mu(n).$$

Our first result is an exact spectral decomposition of this energy through a compact positive integral operator whose kernel is a time-changed Brownian covariance kernel. Our second result is probabilistic: under RH and the negative-moment hypothesis $J_{-1}(T) \ll T$, the process $Y \mapsto \mathcal{M}_L(e^Y)$ has a limiting distribution with an explicit first moment. Under linear independence of the positive zero ordinates, the limit law is identified with a quadratic functional of independent unit-circle phases.

Prior work and positioning

For general background on the Riemann zeta function and the Mertens function we refer to Titchmarsh–Heath-Brown [1]; for conditional estimates for partial sums of the Möbius function,

see Soundararajan [7]. The present work sits at the intersection of three lines of research, and we state at the outset what is taken from each and what is new.

Limiting distributions of linear functionals. The limiting-distribution perspective on prime number error terms originates with Wintner [2] and appears in modern form in Rubinstein–Sarnak [3]. For the Mertens function, Ng [4] proved, assuming RH and $J_{-1}(T) \ll T$, that the linear functional $e^{-y/2}M(e^y)$ possesses a limiting distribution, and under the additional linear independence hypothesis described that distribution through independent phases attached to the zeros. Akbary–Ng–Shahabi [15] placed this in a general framework of B^2 -almost periodic functions with values in $\mathbb{R}^\ell$, and Humphries [14] treated weighted sums of the Liouville function by the same method. Our hypotheses are exactly those of Ng, and the mean-square approximation of $e^{-y/2}M(e^y)$ from [4] is the analytic input of Section 4. What is new is the object: $\mathcal{M}_L(e^Y)$ is a *quadratic* functional of a fixed-length segment of the normalized Mertens process, so the finite-dimensional vector-valued theorem of [15] does not apply directly, and the passage from the B^2 -approximation of the process to the limiting law of its energy (Sections 4.2–4.3) is carried out here in the Wasserstein-1 metric.

Mean squares over fixed-ratio windows. For the prime-counting error, Brent–Platt–Trudgian [22] studied $I(X) = \int_X^{2X} (\psi(x) - x)^2 dx$, proved under RH that $\limsup I(X)/X^2 \leq 0.8603$, and showed that $I(X)/X^2$ has no limit. $\mathcal{M}_L$ is a Möbius analogue of $I(X)/X^2$, with the increment anchored at the left endpoint of the window and weighted by y^{-2} ; the anchoring is what makes the energy a norm of the process F_Y of Section 4. We obtain a limiting distribution rather than bounds on $\limsup$ and $\liminf$, and the two viewpoints are consistent: Corollary 4.3 gives a quantitative version of the non-existence of an ordinary limit, under an independence hypothesis on γ_1 (for computational work on linear relations among zeta ordinates see Best–Trudgian [16]). A direct analogue of the unconditional non-existence argument of [22] is left open (Remark 4.5).

Short intervals. The fixed logarithmic window $[e^{-L}X, X]$ should be distinguished from the classical short-interval problem for Möbius sums, where the interval length h is much smaller than X . The pseudorandom short-interval heuristics go back to Good–Churchhouse [8], whose random-walk model of $M(x)$ is also the origin of the Brownian picture behind the kernel K_L . In unpublished work, Gao [10] proved on RH the mean-square bound $\int_X^{2X} |M(x+h) - M(x)|^2 dx \ll Xh(\log X)^4$ for $h \geq (\log X)^4$, a Möbius analogue of Selberg’s mean-square estimate for primes, and Ng [9] obtained, under a Möbius version of the Hardy–Littlewood prime k -tuples conjecture, a Gaussian limit law for $M(x+h) - M(x)$ with variance $\sim 6h/\pi^2$. (The published paper of Gao–Zhao [11] on mean squares of quadratic twists of μ concerns a different object.) In the fixed-ratio regime $h \asymp X$ considered here the individual zeros are visible, the limit law is non-Gaussian, and its mean is a convergent sum over the zeros.

Kernels. Kernels of min type on Hilbert spaces of functions of μ also appear in the Nyman–Beurling–Báez-Duarte approach to RH, where the relevant object is the multiplicative autocorrelation of the fractional-part function [13]. The kernel K_L of this paper is of a different nature: it is the covariance of a time-changed Brownian motion, its spectral decomposition is a regular Sturm–Liouville problem, and no criterion for RH is attached to it.

We are not aware of prior work on the fixed-ratio quadratic functional $\mathcal{M}_L$, on its spectral decomposition, or on limiting distributions of quadratic functionals of $e^{-y/2}M(e^y)$; the results of Sections 3–4 appear to be new in this combination, while every ingredient (the hypotheses, the B^2 -framework and the Brownian model) is classical and is cited above.

2 Notation and standing assumptions

Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad Z(y) = e^{-y/2}M(e^y).$$

Fix $L > 0$ and write

$$a = e^{-L}.$$

For $X > 0$ define

$$\mathcal{M}_L(X) = \int_{aX}^X \frac{|M(y) - M(aX)|^2}{y^2} dy.$$

When zero sums are used, we write

$$\rho = \frac{1}{2} + i\gamma, \quad c_\gamma = \frac{1}{\rho\zeta'(\rho)}.$$

3 The Brownian-type kernel and its spectral decomposition

3.1 The exact quadratic form

Expanding the square gives

$$\mathcal{M}_L(X) = \sum_{aX < m, n \leq X} \mu(m)\mu(n) \left(\frac{1}{\max(m, n)} - \frac{1}{X} \right).$$

After scaling $s = m/X$ and $t = n/X$,

$$\mathcal{M}_L(X) = \frac{1}{X} \sum_{aX < m, n \leq X} \mu(m)\mu(n) K_L(m/X, n/X)$$

with

$$K_L(s, t) = \frac{1}{\max(s, t)} - 1, \quad a \leq s, t \leq 1.$$

3.2 Positivity and the Green-kernel interpretation

The kernel has the factorization

$$K_L(s, t) = \int_a^1 \frac{\mathbf{1}_{s \leq u} \mathbf{1}_{t \leq u}}{u^2} du.$$

Hence, for arbitrary points $t_1, \dots, t_N \in [a, 1]$ and coefficients $c_1, \dots, c_N$,

$$\sum_{m, n=1}^N c_m \overline{c_n} K_L(t_m, t_n) = \int_a^1 \frac{1}{u^2} \left| \sum_{t_n \leq u} c_n \right|^2 du \geq 0.$$

Thus K_L is continuous and positive semidefinite. Moreover, with

$$\tau(t) = t^{-1} - 1,$$

we have

$$K_L(s, t) = \min(\tau(s), \tau(t)).$$

Hence K_L is exactly the covariance kernel of the time-changed standard Brownian motion $B(\tau(t))$ on $[a, 1]$. This is the reason for the term *Brownian-type kernel*; the Mercer expansion below is the corresponding Karhunen–Loève expansion relative to the measure dt .

Let

$$(T_L f)(t) = \int_a^1 K_L(s, t) f(s) ds.$$

Splitting the integral at $s = t$ gives

$$(T_L f)(t) = \left(\frac{1}{t} - 1\right) \int_a^t f(s) ds + \int_t^1 \left(\frac{1}{s} - 1\right) f(s) ds,$$

and therefore

$$(T_L f)'(t) = -\frac{1}{t^2} \int_a^t f(s) ds.$$

Consequently,

$$-(t^2(T_L f)'(t))' = f(t),$$

with

$$(T_L f)'(a) = 0, \quad (T_L f)(1) = 0.$$

For standard facts on regular self-adjoint Sturm–Liouville problems we use the framework of Zettl [18].

Define

$$\mathcal{L}u = -\frac{d}{dt} (t^2 u'(t))$$

with domain

$$D(\mathcal{L}) = \{u \in H^2([a, 1]) : u'(a) = 0, u(1) = 0\}.$$

Then $T_L : L^2([a, 1]) \rightarrow D(\mathcal{L})$ and

$$\mathcal{L}T_L = I.$$

Hence T_L is injective. In particular, its integral operator is strictly positive even though the kernel is only required to be positive semidefinite for Mercer's theorem.

For fixed $s \in (a, 1)$, $t \mapsto K_L(s, t)$ is constant for $t < s$ and equals $t^{-1} - 1$ for $t > s$. Thus

$$-\frac{d}{dt} \left(t^2 \frac{\partial}{\partial t} K_L(s, t) \right) = \delta_s(t),$$

with the same boundary conditions. Hence K_L is the Green kernel of $\mathcal{L}$ and $T_L = \mathcal{L}^{-1}$.

3.3 A spectral gap for $\mathcal{L}$

Lemma 3.1.

$$\text{spec}(\mathcal{L}) \subset (1/4, \infty).$$

Proof. Integration by parts and the boundary conditions give

$$\langle \mathcal{L}u, u \rangle = \int_a^1 t^2 |u'(t)|^2 dt > 0$$

for nonzero $u \in D(\mathcal{L})$.

Suppose first that an eigenvalue η satisfies $0 < \eta < 1/4$, and write

$$\eta = \frac{1}{4} - \kappa^2, \quad \kappa > 0.$$

The solution of

$$-(t^2 u'(t))' = \eta u(t)$$

satisfying $u(1) = 0$ is a scalar multiple of

$$u(t) = t^{-1/2} \sinh\left(\kappa \log \frac{1}{t}\right).$$

The condition $u'(a) = 0$ would imply

$$\tanh(\kappa L) = -2\kappa,$$

which is impossible for $\kappa > 0$.

If $\eta = 1/4$, then the solution satisfying $u(1) = 0$ is a scalar multiple of

$$t^{-1/2} \log(1/t),$$

whose derivative at a is

$$-a^{-3/2} \left(1 + \frac{L}{2}\right) \neq 0.$$

Thus no eigenvalue lies in $(0, 1/4]$. □

3.4 Eigenvalues and eigenfunctions

Let $T_L e = \lambda e$. Since T_L maps into $D(\mathcal{L})$ and is injective, e is a classical eigenfunction of $\mathcal{L}$ and $\lambda \neq 0$. Applying $\mathcal{L}$ gives

$$-(t^2 e'(t))' = \lambda^{-1} e(t).$$

By the previous lemma,

$$\lambda^{-1} = \nu^2 + \frac{1}{4}$$

for some $\nu > 0$. The differential equation becomes

$$t^2 e'' + 2te' + \left(\nu^2 + \frac{1}{4}\right) e = 0.$$

Its solutions are linear combinations of

$$t^{-1/2} \cos(\nu \log t), \quad t^{-1/2} \sin(\nu \log t).$$

The condition $e(1) = 0$ removes the cosine component, so we may write

$$e_\nu(t) = t^{-1/2} \sin\left(\nu \log \frac{1}{t}\right).$$

The Neumann condition at $a = e^{-L}$ gives

$$\frac{1}{2} \sin(\nu L) + \nu \cos(\nu L) = 0,$$

equivalently

$$\boxed{\tan(\nu L) = -2\nu.}$$

Let

$$0 < \nu_1 < \nu_2 < \dots$$

be the positive roots. Then

$$\boxed{\lambda_j = \frac{1}{\nu_j^2 + 1/4}.}$$

The normalized eigenfunctions are

$$\boxed{e_j(t) = \frac{t^{-1/2} \sin(\nu_j \log(1/t))}{N_j},}$$

where

$$N_j^2 = \int_a^1 t^{-1} \sin^2 \left(\nu_j \log \frac{1}{t} \right) dt.$$

Using

$$\tan(\nu_j L) = -2\nu_j$$

one obtains

$$\sin(2\nu_j L) = -\frac{4\nu_j}{1 + 4\nu_j^2},$$

hence

$$N_j^2 = \frac{L}{2} + \frac{1}{1 + 4\nu_j^2} = \frac{L}{2} + \frac{\lambda_j}{4}.$$

The roots satisfy

$$\nu_j = \frac{(j - \frac{1}{2})\pi}{L} + O(j^{-1}),$$

and consequently

$$\lambda_j \sim \frac{L^2}{\pi^2(j - \frac{1}{2})^2}.$$

Since T_L is compact, self-adjoint and strictly positive, the $\{e_j\}$ form a complete orthonormal basis of $L^2([a, 1])$.

3.5 Mercer expansion and the exact energy decomposition

Since K_L is continuous and positive semidefinite on the compact square $[a, 1]^2$, Mercer's theorem [17] yields

$$K_L(s, t) = \sum_{j=1}^{\infty} \lambda_j e_j(s) e_j(t),$$

with pointwise, in fact uniform, convergence on $[a, 1]^2$.

For fixed X the sum over m, n in the quadratic form is finite. Therefore substitution of the Mercer expansion and interchange with the finite arithmetic sum are immediate. We obtain

$$\mathcal{M}_L(X) = \frac{1}{X} \sum_{j=1}^{\infty} \lambda_j \left| \sum_{aX < n \leq X} \mu(n) e_j(n/X) \right|^2.$$

Define

$$S_j(X) = \frac{1}{\sqrt{X}} \sum_{aX < n \leq X} \mu(n) e_j(n/X).$$

Then

$$\mathcal{M}_L(X) = \sum_{j=1}^{\infty} \lambda_j |S_j(X)|^2.$$

Equivalently,

$$S_j(X) = \frac{1}{N_j} \sum_{aX < n \leq X} \frac{\mu(n)}{\sqrt{n}} \sin \left(\nu_j \log \frac{X}{n} \right).$$

3.6 Trace identity

Mercer's theorem gives

$$\sum_{j=1}^{\infty} \lambda_j = \int_a^1 K_L(t, t) dt.$$

Since $K_L(t, t) = t^{-1} - 1$,

$$\boxed{\sum_{j=1}^{\infty} \frac{1}{\nu_j^2 + 1/4} = L - 1 + e^{-L}.}$$

For

$$L = \log \frac{3}{2}, \quad a = \frac{2}{3},$$

this becomes

$$\sum_{j \geq 1} \lambda_j = \log \frac{3}{2} - \frac{1}{3} = 0.072131774 \dots$$

Moreover,

$$\nu_1 = 4.168481379 \dots, \quad \lambda_1 = 0.056733607 \dots,$$

so

$$\frac{\lambda_1}{\sum_j \lambda_j} = 0.7865 \dots$$

This operator-trace proportion should not be confused with the arithmetic fraction of $\mathcal{M}_L(X)$ or of its limiting mean carried by the first mode; see Section 5.3.

3.7 Coupling a zeta zero to the Brownian modes

For a nontrivial zero

$$\rho = \frac{1}{2} + i\gamma$$

define

$$\boxed{h_j(\rho) = \int_a^1 t^{\rho-1} e_j(t) dt.}$$

In the variable $u = \log(1/t)$,

$$h_j(\rho) = \frac{1}{N_j} \int_0^L e^{-i\gamma u} \sin(\nu_j u) du.$$

By the spectral expansion of T_L ,

$$\sum_{j \geq 1} \lambda_j |h_j(\rho)|^2 = \langle T_L t^{\rho-1}, t^{\rho-1} \rangle.$$

The factorization of the kernel gives

$$\boxed{\sum_{j \geq 1} \lambda_j |h_j(\rho)|^2 = \frac{A_L(\gamma)}{|\rho|^2},}$$

where

$$\boxed{A_L(\gamma) = L + 1 - e^{-L} - 2\Re \frac{1 - e^{(-1/2+i\gamma)L}}{1/2 - i\gamma}.}$$

Define

$$p_j(\gamma) = \frac{\lambda_j |h_j(\rho)|^2 |\rho|^2}{A_L(\gamma)}.$$

Then

$$p_j(\gamma) \geq 0, \quad \sum_{j \geq 1} p_j(\gamma) = 1.$$

4 Limiting distribution of the local Mertens energy

Throughout this section we assume RH and the Gonek–Hejhal type bound [5, 6]

$$J_{-1}(T) = \sum_{0 < \gamma \leq T} \frac{1}{|\zeta'(\rho)|^2} \ll T.$$

In particular, the nontrivial zeros are simple.

For $Y \in \mathbb{R}$ and $0 \leq v \leq L$, put

$$F_Y(v) = Z(Y - L + v) - e^{-v/2} Z(Y - L).$$

We use the convention $Z(y) = 0$ for $y < 0$; equivalently, all long-time averages below may be started at $Y = L$. Direct substitution gives

$$\mathcal{M}_L(e^Y) = \|F_Y\|_{L^2(0,L)}^2.$$

For $T \geq 2$ define

$$Z_T(y) = 2\Re \sum_{0 < \gamma \leq T} c_\gamma e^{i\gamma y}$$

and

$$F_{Y,T}(v) = Z_T(Y - L + v) - e^{-v/2} Z_T(Y - L).$$

Finally,

$$\mathcal{M}_{L,T}(e^Y) = \|F_{Y,T}\|_2^2.$$

Theorem 4.1 (Limiting distribution and first moment). *Assume RH and*

$$J_{-1}(T) \ll T.$$

Then $Y \mapsto \mathcal{M}_L(e^Y)$ possesses a limiting distribution

$$\Pi_L \in \mathcal{P}_1(\mathbb{R}),$$

and

$$\int_{\mathbb{R}} x d\Pi_L(x) = 2 \sum_{\gamma > 0} \frac{A_L(\gamma)}{|\rho \zeta'(\rho)|^2}.$$

Equivalently, the same quantity is the ordinary time mean of $\mathcal{M}_L(e^Y)$ in the variable $Y = \log X$.

We prove Theorem 4.1 in the next three subsections.

4.1 Mean-square approximation

Ng's explicit formula also contains the constant term -2 and the contribution of the trivial zeros. After normalization by $e^{y/2}$ these terms tend exponentially to zero and hence do not affect the Besicovitch seminorms used below.

Ng's Lemma 10 [4], together with his Lemma 5 which removes the restriction to the special sequence of truncation heights, yields

$$\int_{\log 2}^R |Z(Y) - Z_T(Y)|^2 dY \ll R \frac{\log T}{T^{1/4}} + 1$$

for fixed $T \geq 2$.

Let

$$R_T(Y) = Z(Y) - Z_T(Y).$$

Then

$$F_Y(v) - F_{Y,T}(v) = R_T(Y - L + v) - e^{-v/2} R_T(Y - L),$$

and therefore

$$\|F_Y - F_{Y,T}\|_2^2 \leq 2 \int_0^L |R_T(Y - L + v)|^2 dv + 2(1 - e^{-L}) |R_T(Y - L)|^2.$$

A fixed translation does not alter the B^2 seminorm, since the ratio of the translated averaging length to the original averaging length tends to one. Hence

$$\boxed{\|F - F_T\|_{B^2(L^2(0,L))}^2 \leq C_L \|Z - Z_T\|_{B^2}^2},$$

where

$$\boxed{C_L = 2(L + 1 - e^{-L})}.$$

Thus

$$\|F - F_T\|_{B^2(L^2)} \longrightarrow 0.$$

Moreover,

$$\sup_T \|Z_T\|_{B^2}^2 = 2 \sup_T \sum_{0 < \gamma \leq T} |c_\gamma|^2 \leq \beta_M,$$

where

$$\beta_M = 2 \sum_{\gamma > 0} \frac{1}{|\rho \zeta'(\rho)|^2} < \infty.$$

Consequently

$$\sup_T \|F_T\|_{B^2(L^2)}^2 \leq C_L \beta_M,$$

and the same bound holds for F , since $\|Z\|_{B^2} \leq \|Z_T\|_{B^2} + \|Z - Z_T\|_{B^2}$ for every T and the last term tends to zero.

4.2 Passage from the Hilbert-valued process to the energy

For arbitrary vectors u, v in a Hilbert space,

$$|\|u\|^2 - \|v\|^2| \leq \|u - v\|(\|u\| + \|v\|).$$

Therefore

$$|\mathcal{M}_L(e^Y) - \mathcal{M}_{L,T}(e^Y)| \leq \|F_Y - F_{Y,T}\|_2 (\|F_Y\|_2 + \|F_{Y,T}\|_2).$$

By Cauchy–Schwarz,

$$\|\mathcal{M}_L - \mathcal{M}_{L,T}\|_{B^1} \leq \|F - F_T\|_{B^2(L^2)} \left(\limsup_{R \rightarrow \infty} \frac{1}{R} \int_0^R (\|F_Y\|_2 + \|F_{Y,T}\|_2)^2 dY \right)^{1/2}.$$

The second factor is uniformly bounded:

$$\limsup_{R \rightarrow \infty} \frac{1}{R} \int_0^R (\|F_Y\|_2 + \|F_{Y,T}\|_2)^2 dY \leq 4C_L \beta_M.$$

Hence

$$\boxed{\|\mathcal{M}_L - \mathcal{M}_{L,T}\|_{B^1} \rightarrow 0.}$$

4.3 Finite-dimensional laws and Wasserstein convergence

Fix T and let

$$0 < \gamma_1 < \dots < \gamma_N \leq T.$$

The function $\mathcal{M}_{L,T}(e^Y)$ is a continuous function of the phase vector

$$(e^{i\gamma_1 Y}, \dots, e^{i\gamma_N Y}).$$

Let

$$\mathbb{G}_T := \text{cl}\{(e^{i\gamma_1 Y}, \dots, e^{i\gamma_N Y}) : Y \in \mathbb{R}\} \subseteq \mathbb{T}^N.$$

By Kronecker–Weyl, the orbit is equidistributed in this compact subtorus with respect to normalized Haar measure. Thus $\mathcal{M}_{L,T}(e^Y)$ has a limiting distribution $\Pi_{L,T}$, without any linear independence assumption.

For a 1-Lipschitz function h normalized by $h(0) = 0$, the time average defining $\int h d\Pi_{L,T}$ is legitimate even though h need not be bounded, because $\mathcal{M}_{L,T}$ is bounded for fixed T .

$$\left| \int h d\Pi_{L,T} - \int h d\Pi_{L,S} \right| \leq \|\mathcal{M}_{L,T} - \mathcal{M}_{L,S}\|_{B^1}.$$

Kantorovich–Rubinstein duality [19, Theorem 5.10(i) and Remark 6.5] therefore gives

$$\boxed{W_1(\Pi_{L,T}, \Pi_{L,S}) \leq \|\mathcal{M}_{L,T} - \mathcal{M}_{L,S}\|_{B^1}.$$

Hence $\{\Pi_{L,T}\}$ is Cauchy in $(\mathcal{P}_1(\mathbb{R}), W_1)$. By the completeness theorem for Wasserstein spaces [19, Theorem 6.18], there exists

$$\Pi_L \in \mathcal{P}_1(\mathbb{R})$$

such that

$$\boxed{W_1(\Pi_{L,T}, \Pi_L) \rightarrow 0.}$$

For every bounded Lipschitz function h ,

$$\begin{aligned} \limsup_{R \rightarrow \infty} \left| \frac{1}{R} \int_0^R h(\mathcal{M}_L(e^Y)) dY - \int h d\Pi_L \right| &\leq \text{Lip}(h) \|\mathcal{M}_L - \mathcal{M}_{L,T}\|_{B^1} \\ &\quad + \limsup_{R \rightarrow \infty} \left| \frac{1}{R} \int_0^R h(\mathcal{M}_{L,T}(e^Y)) dY - \int h d\Pi_{L,T} \right| \\ &\quad + \text{Lip}(h) W_1(\Pi_{L,T}, \Pi_L). \end{aligned}$$

The middle term is zero for fixed T , while the first and third terms tend to zero with T . Thus $\mathcal{M}_L(e^Y)$ possesses limiting distribution Π_L . The bounded-Lipschitz formulation is equivalent to the usual bounded-continuous formulation of weak convergence on $\mathbb{R}$.

4.4 The first moment

For fixed T , all nonzero frequencies average to zero. No LI hypothesis is needed: if $\gamma \neq \gamma'$, then $\gamma - \gamma' \neq 0$, while $\gamma + \gamma' > 0$. Hence

$$\int x d\Pi_{L,T}(x) = 2 \sum_{0 < \gamma \leq T} \frac{A_L(\gamma)}{|\rho\zeta'(\rho)|^2}.$$

Since $\Pi_{L,T} \rightarrow \Pi_L$ in W_1 , the first moments converge, and

$$\boxed{\int x d\Pi_L(x) = 2 \sum_{\gamma > 0} \frac{A_L(\gamma)}{|\rho\zeta'(\rho)|^2}.$$

The same value equals the time mean:

$$\boxed{\lim_{R \rightarrow \infty} \frac{1}{R} \int_0^R \mathcal{M}_L(e^Y) dY = \int x d\Pi_L(x).$$

For

$$L = \log \frac{3}{2}$$

we have $1 - e^{-L} = 1/3$. Thus

$$A_L(\gamma) = L + \frac{1}{3} + O(\gamma^{-1}).$$

If one additionally assumes Gonek's conjectural asymptotic [5]

$$J_{-1}(T) \sim \frac{3}{\pi^3} T,$$

then partial summation yields

$$\sum_{\gamma > T} \frac{2A_L(\gamma)}{|\rho\zeta'(\rho)|^2} \sim \frac{6(L + 1 - e^{-L})}{\pi^3 T}.$$

At $L = \log(3/2)$ the coefficient equals

$$\frac{6(\log(3/2) + 1/3)}{\pi^3} = 0.1429 \dots,$$

so the predicted truncation tail is asymptotic to

$$\frac{0.143}{T}.$$

4.5 Identification under linear independence

Theorem 4.2 (Random-phase identification under LI). *Under the assumptions of Theorem 4.1, assume additionally that the positive ordinates $\{\gamma\}$ are linearly independent over $\mathbb{Q}$. Then Π_L is the law of*

$$\mathcal{E}_L = \|G_L\|_{L^2(0,L)}^2,$$

where

$$G_L(v) = 2\Re \sum_{\gamma > 0} \frac{e^{i\gamma v} - e^{-v/2}}{\rho\zeta'(\rho)} U_\gamma$$

and the U_γ are independent Haar-uniform variables on the unit circle. The random series is well defined in $L^2(\Omega; L^2(0, L))$ and almost surely.

Proof. Under LI, for every T the subtorus $\mathbb{G}_T$ of Section 4.3 is the full torus $\mathbb{T}^N$ and its Haar measure is the product Haar measure. Hence $\Pi_{L,T}$ is the law of the truncated random energy

$$\mathcal{E}_{L,T} = \left\| \sum_{0 < \gamma \leq T} X_\gamma \right\|_{L^2(0,L)}^2, \quad X_\gamma(v) = 2\Re \left(\frac{e^{i\gamma v} - e^{-v/2}}{\rho \zeta'(\rho)} U_\gamma \right).$$

The summands X_γ are independent, centered random variables with values in $L^2(0,L)$, and

$$\mathbb{E} \|X_\gamma\|_2^2 = \frac{2A_L(\gamma)}{|\rho \zeta'(\rho)|^2}.$$

The sum of these second moments converges. Hence the random series $\sum_\gamma X_\gamma$ converges in $L^2(\Omega; L^2(0,L))$ and almost surely, e.g. by the standard convergence theory for independent Banach-space-valued random series; see Itô–Nisio [20] or Ledoux–Talagrand [21]. Its sum is G_L .

By the Hilbert-space inequality of Section 4.2 and Cauchy–Schwarz, the truncated random energies $\mathcal{E}_{L,T}$ converge to $\mathcal{E}_L$ in $L^1(\Omega)$, hence $W_1(\text{Law}(\mathcal{E}_{L,T}), \text{Law}(\mathcal{E}_L)) \rightarrow 0$. Since also $W_1(\Pi_{L,T}, \Pi_L) \rightarrow 0$, we conclude

$$\Pi_L = \text{Law}(\mathcal{E}_L). \quad \square$$

4.6 Individual Brownian modes

Since $e_j(1) = 0$, summation by parts gives

$$S_j(e^Y) = - \int_0^L F_Y(v) (ae^v)^{3/2} e'_j(ae^v) dv.$$

Thus S_j is a bounded linear functional of F_Y . The B^2 approximation transfers to every fixed S_j and every finite vector $(S_1, \dots, S_J)$. This is also consistent with the finite-dimensional vector-valued framework of Akbary–Ng–Shahabi [15, Theorem 1.4].

For fixed j , integration by parts in the definition of h_j gives

$$|h_j(\rho)| \ll_j \frac{1}{|\gamma|}.$$

Thus

$$\sum_{\gamma > 0} \frac{|h_j(\rho)|^2}{|\zeta'(\rho)|^2}$$

converges under the standing hypothesis.

The truncated explicit formula for the j th mode is

$$S_{j,T}(e^Y) = 2\Re \sum_{0 < \gamma \leq T} \frac{h_j(\rho)}{\zeta'(\rho)} e^{i\gamma Y}.$$

Therefore, without LI,

$$\lim_{R \rightarrow \infty} \frac{1}{R} \int_0^R \lambda_j |S_j(e^Y)|^2 dY = 2\lambda_j \sum_{\gamma > 0} \frac{|h_j(\rho)|^2}{|\zeta'(\rho)|^2}.$$

4.7 High-zero modal weights and escape of mass

For fixed j ,

$$h_j(\rho) = \frac{1}{N_j} \int_0^L e^{-i\gamma u} \sin(\nu_j u) du.$$

Integration by parts yields

$$\gamma^2 |h_j(\rho)|^2 \longrightarrow \frac{\sin^2(\nu_j L)}{N_j^2}.$$

Since

$$\sin^2(\nu_j L) = \nu_j^2 \lambda_j$$

and

$$A_L(\gamma) \longrightarrow L + 1 - e^{-L},$$

we obtain

$$p_j(\infty) = \frac{\lambda_j^2 \nu_j^2}{N_j^2 (L + 1 - e^{-L})}.$$

The total mass retained by all fixed modes has an exact closed form. Indeed, Mercer's identity at (a, a) gives

$$K_L(a, a) = \frac{1}{a} - 1 = \sum_{j \geq 1} \lambda_j e_j(a)^2.$$

Because

$$e_j(a)^2 = \frac{\sin^2(\nu_j L)}{aN_j^2} = \frac{\nu_j^2 \lambda_j}{aN_j^2},$$

we obtain

$$\sum_{j \geq 1} \frac{\lambda_j^2 \nu_j^2}{N_j^2} = 1 - a = 1 - e^{-L}.$$

Consequently,

$$\sum_{j \geq 1} p_j(\infty) = \frac{1 - e^{-L}}{L + 1 - e^{-L}}.$$

The escaping mass is therefore exactly

$$1 - \sum_{j \geq 1} p_j(\infty) = \frac{L}{L + 1 - e^{-L}}.$$

For $L = \log(3/2)$,

$$\sum_j p_j(\infty) = 0.45118\dots, \quad 1 - \sum_j p_j(\infty) = 0.54882\dots$$

This has a useful interpretation. In the high-zero limit,

$$A_L(\gamma) \longrightarrow L + (1 - e^{-L}).$$

The term $1 - e^{-L}$ comes from

$$\int_0^L e^{-v} dv,$$

that is, from the smooth anchor $e^{-v/2}Z(Y-L)$, and it is responsible for the nonvanishing mass retained by fixed low Brownian modes. The term L comes from

$$\int_0^L |e^{i\gamma v}|^2 dv.$$

The formula above proves that this portion of the mass escapes every fixed finite collection of modes as $\gamma \rightarrow \infty$. The stronger statement that it is concentrated on modes satisfying $\nu_j \asymp \gamma$ is a natural resonance heuristic, but is not asserted here without a uniform two-parameter estimate for $h_j(\rho)$.

4.8 A quantitative oscillation corollary

Let γ_1 denote the smallest positive ordinate and define

$$B_1 = \int_0^L \left(e^{i\gamma_1 v} - e^{-v/2} \right)^2 dv.$$

Consider the hypothesis

$$\boxed{\gamma_1 \notin \text{span}_{\mathbb{Q}}\{\gamma_k : k \geq 2\}.}$$

This is weaker than full linear independence; for computational work on linear relations among zeta ordinates see Best–Trudgian [16].

Under this assumption, for every finite truncation containing γ_1 , the orbit closure splits as the product of the γ_1 circle and a subtorus generated by the remaining ordinates. Conditional on all other phases, the truncated energy is a real trigonometric polynomial $f_T(\phi_1)$ whose Fourier coefficient at frequency 2 is exactly

$$c_{\gamma_1}^2 B_1.$$

Cross terms involving any other zero contribute only frequencies ± 1 in ϕ_1 .

For any constant $c \in \mathbb{R}$,

$$|c_{\gamma_1}^2 B_1| = \left| \int_0^{2\pi} (f_T(\phi) - c) e^{-2i\phi} \frac{d\phi}{2\pi} \right| \leq \int_0^{2\pi} |f_T(\phi) - c| \frac{d\phi}{2\pi}.$$

Averaging over the remaining torus gives

$$\int_{\mathbb{R}} |x - c| d\Pi_{L,T}(x) \geq |c_{\gamma_1}|^2 |B_1|$$

for every truncation T containing the first zero and every $c \in \mathbb{R}$.

Since $x \mapsto |x - c|$ is 1-Lipschitz, W_1 convergence yields

$$\boxed{\int_{\mathbb{R}} |x - c| d\Pi_L(x) \geq |c_{\gamma_1}|^2 |B_1| > 0 \quad (c \in \mathbb{R}).}$$

In particular Π_L is nondegenerate.

Corollary 4.3 (Quantitative oscillation). *Assume RH,*

$$J_{-1}(T) \ll T,$$

and

$$\gamma_1 \notin \text{span}_{\mathbb{Q}}\{\gamma_k : k \geq 2\}.$$

Then $\mathcal{M}_L(e^Y)$ has no finite ordinary limit as $Y \rightarrow \infty$. More precisely,

$$\boxed{\limsup_{Y \rightarrow \infty} \mathcal{M}_L(e^Y) - \liminf_{Y \rightarrow \infty} \mathcal{M}_L(e^Y) \geq 2|c_{\gamma_1}|^2 |B_1|.$$

Proof. If the limsup is infinite, the conclusion is immediate. Otherwise let

$$\ell_- = \liminf_{Y \rightarrow \infty} \mathcal{M}_L(e^Y), \quad \ell_+ = \limsup_{Y \rightarrow \infty} \mathcal{M}_L(e^Y).$$

Every limiting distribution of long-time averages is supported in $[\ell_-, \ell_+]$. Taking $c = (\ell_- + \ell_+)/2$ gives

$$\int |x - c| d\Pi_L(x) \leq \frac{\ell_+ - \ell_-}{2}.$$

Combine this with the preceding lower bound. $\square$

For $L = \log(3/2)$, numerically

$$|c_{\gamma_1}|^2 = 0.007946\dots, \quad |B_1| = 0.3616\dots,$$

hence

$$\boxed{\limsup \mathcal{M}_L(e^Y) - \liminf \mathcal{M}_L(e^Y) \geq 0.0057\dots}$$

Remark 4.4. The proof above avoids the difficulty that nondegeneracy of each truncated law need not survive weak or W_1 convergence. The fixed Fourier coefficient at $2\phi_1$ gives a truncation-independent lower bound in a 1-Lipschitz observable, and therefore passes directly to the W_1 limit.

Remark 4.5. Without any independence assumption on γ_1 , a direct analogue of the argument of Brent–Platt–Trudgian [22] would require additional control of the truncation error at specially chosen individual times. The present B^2 argument controls moving-window averages but does not control the anchored point value $Z(Y - L)$ strongly enough for that purpose. We leave the unconditional removal of the frequency independence hypothesis as an open refinement.

5 Numerical illustrations

The numerical experiments in this section are checks of agreement between the arithmetic data and the explicit-formula description; they are not evidence for RH or for the auxiliary hypotheses of Section 4, and they prove nothing. Unless stated otherwise we use

$$L = \log(3/2), \quad a = 2/3.$$

All code and data are available at <https://github.com/klukaspl/local-mertens-energy>. The Möbius function and $M(n)$ were computed by a sieve up to 10^8 . We used 30,000 sample points approximately uniform in

$$Y = \log X \in [\log 10^4, \log 10^8] = [9.21034\dots, 18.42068\dots].$$

For integer sample points X , the energy was evaluated from the exact step-function identity

$$\mathcal{M}_L(X) = \sum_{n=\lfloor aX \rfloor + 1}^{X-1} (M(n) - M(\lfloor aX \rfloor))^2 \left(\frac{1}{n} - \frac{1}{n+1} \right),$$

so no numerical quadrature is involved.

The full Y -range contains only about 20.7 periods of the phase $e^{i\gamma_1 Y}$ (and about 41.4 periods of its self-energy harmonic $e^{2i\gamma_1 Y}$). The 30,000 sampled values are therefore highly correlated; numerical means below should not be read as four-digit statistical estimates.

5.1 Trajectory reconstruction from the zeros

Figure 1 compares the empirical trajectory with truncations of the explicit formula using the first 1, 10, 80 and 1000 positive zeros. The displayed panel is the final five logarithmic units of the full sampled range. The truncated energy $\mathcal{M}_{L,T}(e^Y)$ is evaluated in closed form from the kernels $\int_0^L g_\gamma \overline{g_{\gamma'}} dv$ and $\int_0^L g_\gamma g_{\gamma'} dv$, where $g_\gamma(v) = e^{i\gamma v} - e^{-v/2}$, and the implementation is checked against direct quadrature. The statistics below are computed on a subsample of 6000 points covering the full range $Y \in [\log 10^4, \log 10^8]$:

number of zeros N	γ_N	correlation	RMS error
1	14.13	0.3077	0.01546
10	49.77	0.8097	0.00800
80	201.26	0.9624	0.00352
300	541.85	0.9851	0.00220
1000	1419.42	0.9947	0.00131

Between $N = 80$ and $N = 1000$ the RMS error decreases approximately like $\gamma_N^{-1/2}$, as one expects when the residual is dominated by cross terms between the retained zeros and a tail of mean size $\asymp 1/\gamma_N$.

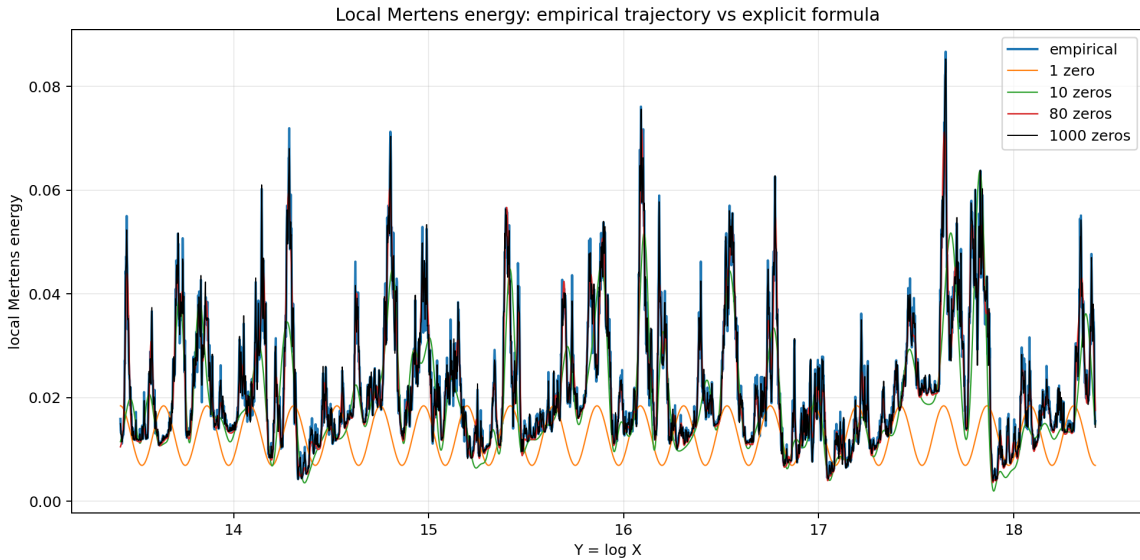


Figure 1: Empirical local Mertens energy versus explicit-formula reconstructions using the first 1, 10, 80 and 1000 positive zeros. The horizontal variable is $Y = \log X$; only the last five log-units of the computed interval are shown for readability.

5.2 Mean value: arithmetic data and zero sums

The empirical means over logarithmic tails of the sampled range are:

range	empirical mean of $\mathcal{M}_L(X)$
$10^4 \leq X \leq 10^8$	0.02230
$10^5 \leq X \leq 10^8$	0.02189
$10^6 \leq X \leq 10^8$	0.02212
$10^7 \leq X \leq 10^8$	0.02154

Their variation is consistent with the small number of effective low-frequency periods described above.

For comparison, the partial sums of Ng's constant are

$$\beta_{M,80} = 2 \sum_{k \leq 80} |\rho_k \zeta'(\rho_k)|^{-2} = 0.0281400\dots, \quad \beta_{M,1000} = 0.0289079\dots$$

Table 1 shows the partial zero sum for the limiting mean and, for orientation, the conjectural Gonek tail correction $0.143/\gamma_N$.

N	γ_N	partial mean	$0.143/\gamma_N$	corrected	$J_{-1}(\gamma_N)/\gamma_N$
10	49.774	0.019462	0.002872	0.022334	0.1190
20	77.145	0.020468	0.001853	0.022321	0.1137
40	122.947	0.021139	0.001163	0.022302	0.1087
80	201.265	0.021557	0.000710	0.022267	0.1021
120	269.970	0.021728	0.000530	0.022257	0.0993
160	334.211	0.021822	0.000428	0.022250	0.0976
200	396.382	0.021882	0.000361	0.022243	0.0962
300	541.847	0.021970	0.000264	0.022234	0.0944
500	811.184	0.022051	0.000176	0.022227	0.0930
1000	1419.422	0.022124	0.000101	0.022225	0.0950

Table 1: Partial zero sums for the predicted limiting mean. The fourth column uses the conjectural asymptotic tail from Gonek's negative-moment conjecture; it is not used in the proof of Theorem 4.1. The last column may be compared with the conjectural limit $3/\pi^3 = 0.09675\dots$

The corrected column stabilizes near 0.02222 once several hundred zeros are included, and we record

$$\mathbb{E}_{\Pi_L} X \approx 0.0222,$$

with the caveat that the tail correction itself is conjectural. The empirical means above, and the full-sample empirical mean 0.02230, are consistent with this value within the accuracy permitted by the short Y -range. Figure 2 shows the same comparison graphically.

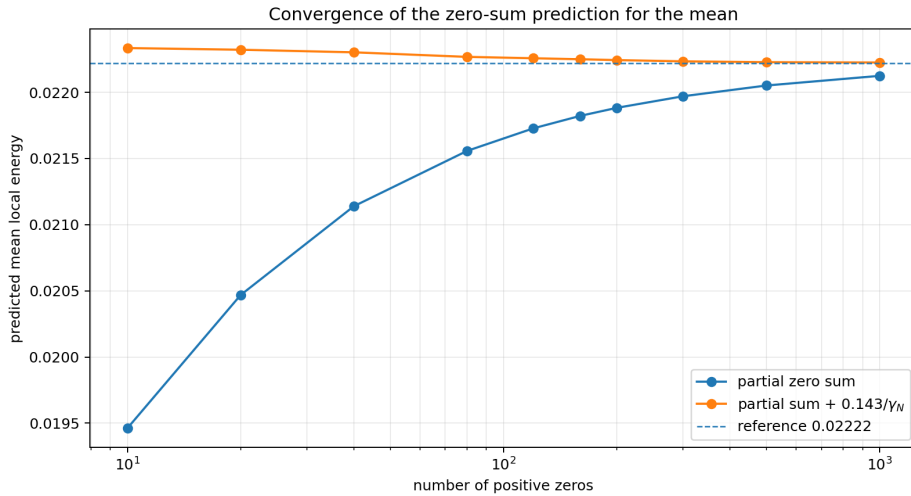


Figure 2: Partial zero sums for the predicted mean, with and without the conjectural asymptotic tail correction $0.143/\gamma_N$.

5.3 Three different notions of dominance

Three percentages that arise naturally in the paper measure different phenomena:

quantity	proportion
first eigenvalue in $\text{tr } T_L$	78.65%
first zeta zero in the predicted limiting mean	$\approx 57\%$
first Brownian mode in the predicted limiting mean	$\approx 46.6\%$

Numerically, the first zero contributes

$$2A_L(\gamma_1)|c_{\gamma_1}|^2 = 0.0126788\dots$$

to a limiting mean near 0.0222.

For the first Brownian mode, Section 4.6 gives the limiting mean of $\lambda_1|S_1|^2$ as a zero sum. Its partial sums are

$$2\lambda_1 \sum_{k \leq N} \frac{|h_1(\rho_k)|^2}{|\zeta'(\rho_k)|^2} = \begin{cases} 0.0101228\dots, & N = 80, \\ 0.0103207\dots, & N = 1000, \end{cases}$$

that is, 46.96% and 46.65% of the corresponding partial means. The omitted zeros also feed the first mode: by Section 4.7, a high zero assigns the asymptotic fraction $p_1(\infty) = 0.349\dots$ of its contribution to the first mode. Including the conjectural tail in both numerator and denominator gives a first-mode share of about 46.6% of the limiting mean, stable between $N = 80$ and $N = 1000$. The first-mode share thus lies between the first-zero value $p_1(\gamma_1) = 55.1\%$ and the high-zero value 34.9%.

The operator identities provide two useful independent checks:

$$\sum_j \lambda_j = \log(3/2) - 1/3 = 0.072131774\dots,$$

and

$$\sum_j p_j(\infty) = \frac{1/3}{\log(3/2) + 1/3} = 0.45118\dots$$

The latter leaves the exact escaping mass 0.54882... derived in Section 4.7.

5.4 First-zero scale and the empirical distribution

For the first zero,

$$A_L(\gamma_1) = 0.79778697\dots, \quad B_1 = 0.36073096\dots - 0.02575568\dots i,$$

and

$$\frac{|B_1|}{A_L(\gamma_1)} = 0.45331557\dots$$

The one-zero random energy therefore has an arcsine law supported on approximately

$$[0.00693, 0.01843].$$

Figure 3 places this scale against the empirical histogram. The full empirical distribution is unimodal and right-skewed; the one-zero arcsine law should therefore be viewed only as the isolated self-interaction of the first zero. Cross terms with the remaining zeros can both increase and decrease the quadratic energy.

The figure also shows 10^5 samples of the random-phase energy of Theorem 4.2, truncated to the first 300 zeros. The sample statistics are close to the arithmetic ones:

	mean	median	standard deviation
arithmetic data, $10^4 \leq X \leq 10^8$	0.02230	0.01850	0.01276
random-phase model, 300 zeros	0.02198	0.01838	0.01303

The model mean agrees with the exact 300-zero partial sum 0.021970 of Table 1.

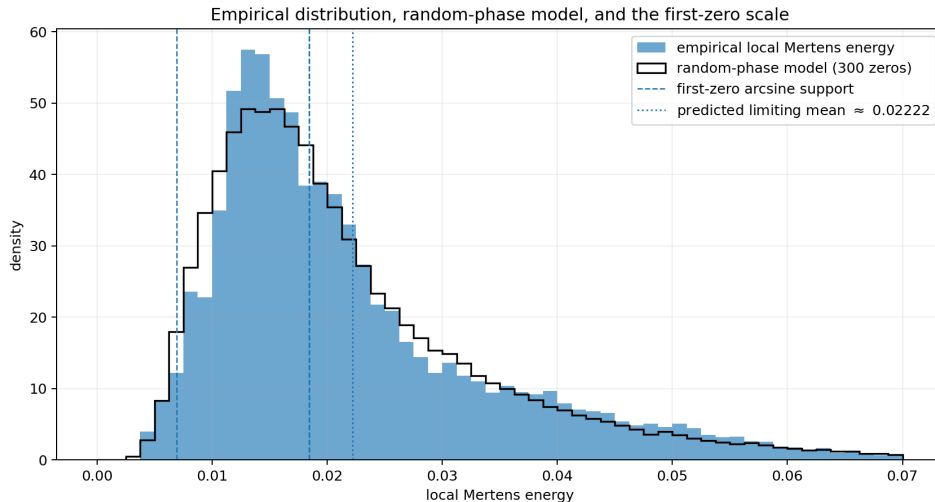


Figure 3: Empirical distribution of the local Mertens energy (filled) and the random-phase model with 300 zeros (outline). The dashed lines indicate the support of the isolated one-zero arcsine contribution; the dotted line marks the predicted limiting mean ≈ 0.0222 . Because the arithmetic samples are strongly correlated in Y , the comparison is descriptive only.

5.5 How individual zeros populate the Brownian modes

The quantities $p_j(\gamma_k)$ describe the fraction of the contribution of a single zero ρ_k carried by Brownian mode j . Figure 4 shows the first five zeros and modes. In particular, the first zero is not resonant with the first Brownian mode: its mass is split principally between modes 1 and 2. The large value of $p_1(\gamma_1)$ is therefore a low-pass effect of λ_1 , rather than direct frequency matching.

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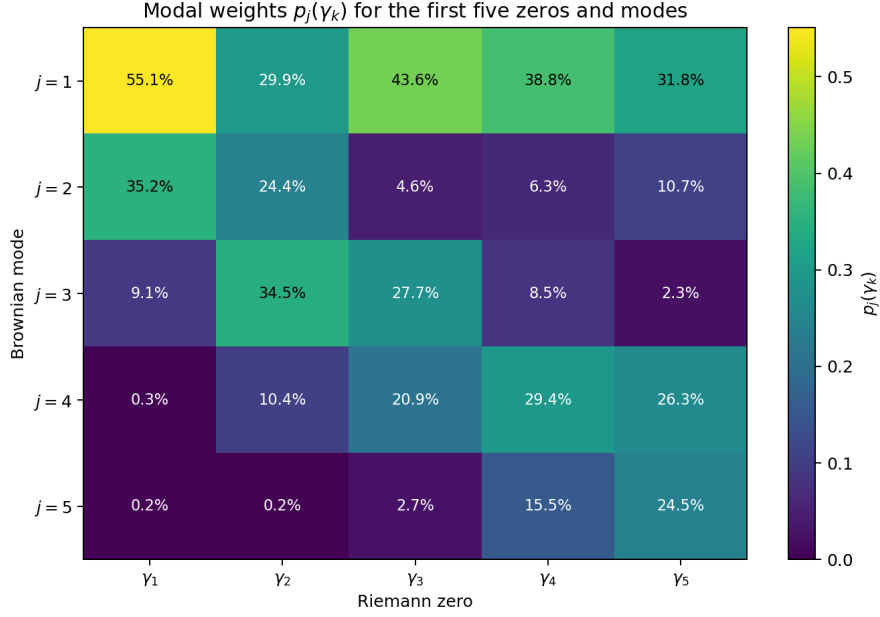


Figure 4: Modal weights $p_j(\gamma_k)$ for the first five positive Riemann zeros and the first five Brownian modes. Each column is part of a probability distribution over all modes.

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